FECON 490 / ECON 690: Business Economics

Details of the Top Down Profit Report Exercise

Department of Economics, Duke University | Spring 2027

This course centers on a semester-long experiential activity: each team operates as a business-economics consultancy advising clients on the next-twelve-month (NTM) U.S. economic outlook and the implications for business profitability. Your team's cumulative product is the *Top Down Profit Report* (TDPR). The client audience has business acumen, but should not be assumed to have specialized expertise in macroeconomic modeling, econometrics, or machine learning.

The objective is not simply to announce a forecast. It is to develop, quantify, test, communicate, and defend a structured outlook that connects economic activity to the economics of real businesses. Your work should make clear where the forecast follows from data and empirical models, where judgment enters, and how uncertainty affects decisions. You are evaluated on the quality of the research process and argument, not on whether an uncertain future outcome agrees with your prediction.

Throughout the semester your team should follow a repeated research cycle:

Frame a question $\longrightarrow$ Collect and analyze data $\longrightarrow$ Estimate and evaluate models $\longrightarrow$ Form a forecast and scenarios $\longrightarrow$ Present and defend $\longrightarrow$ Revise

As in professional research, a well-executed investigation may conclude that a proposed model adds little value and should not be used. That can be strong research if the method, evidence, and conclusion are sound.

Project Structure

The project contains four connected activities:

1. **Three one-slide updates:** All teams briefly present their evolving TDPR on February 17, February 24, and March 3; each is worth two course points.
2. **One full-deck presentation:** Each of the three teams leads an extended, graded in-class discussion on its assigned date: Team A March 17, Team B March 24, Team C March 31.
3. **An ungraded April 7 discussion:** All teams make one-slide updates as an in-class activity. There are no presentation points for this date.
4. **Final comprehensive TDPR:** A cumulative recorded presentation, slide deck, reproducible code and data, and supporting documentation, due according to Canvas. ECON 690 students additionally complete an individual Graduate Research Extension.

These are successive versions of a common research program, not unrelated assignments. You should reuse, correct, and improve valid earlier work as new information and better analysis become available. The final submission should synthesize the work developed during the semester.

Analytical Mandate

The top-down hierarchy

A useful TDPR begins with a coherent forecast for the U.S. macroeconomy and then traces its economic implications through a hierarchy:

Macro $\longrightarrow$ National Income $\longrightarrow$ Industry $\longrightarrow$ Company.

In practical terms, a macro forecast affects expected national sales, input costs, labor costs, financing costs, and profits; those effects differ across industries; and each company faces additional firm-specific factors.

The framework provides a common structure for making projections, discussing alternative scenarios, and attributing deviations from a baseline. The associated lecture notes, *Profit from the Top Down*, develop the theoretical and empirical intuition.

1. **Macro:** Provide NTM numerical expectations and supporting logic for five main concepts: real output growth, labor-market conditions (especially the unemployment rate), inflation (with PCE as the course's main price index), interest rates, and the real broad U.S. dollar. Use consistent measurement, timing, and definitions.
2. **National Income:** Translate the macro outlook into a national income statement covering output/revenues, intermediate inputs/COGS, gross value added, labor and other costs, depreciation, interest where relevant, and profitability. Use the accounting relationships emphasized in the lecture material.
3. **Industry:** Explain why selected industries may respond differently from national activity. Decompose the contribution of aggregate forces and industry-specific factors to revenue, costs, and profit outlooks.
4. **Company:** In the final TDPR, connect the industry outlook to one representative or strategically relevant company in each selected industry, while acknowledging company-specific circumstances.
5. **Risk:** Organize the outlook around at least three plausible joint economic scenarios. Each scenario should include a coherent narrative and probability, as well as implications through all required layers of the hierarchy.

Model-based evidence and judgment

The project combines economics, econometrics, and computation. Teams are expected to **estimate empirical relationships** relevant to the TDPR, particularly between macro variables and national-income components, between national and industry outcomes, or between industry and company outcomes. Merely citing third-party numerical forecasts without analysis does not satisfy the empirical research expectation.

For each material empirical model, be prepared to explain the question, dependent variable and predictors, data source and time period, specification, estimation approach, diagnostics or validation strategy, results, limitations, and relevance to the NTM forecast. Appropriate work might include linear regression; panel or hierarchical specifications; lagged relationships; time-series analysis; regularization; tree-based methods; or other relevant ML models. There is **no required list or count of ML algorithms**. A simple well-validated model is preferable to a complicated poorly justified one. R or Python may be used.

The theoretical hierarchy is primarily a tool for economic interpretation. As discussed in the lecture notes, estimating separate links and then embedding fitted values in subsequent regressions may compound estimation error; pooled or panel-based specifications may be more useful. The macro block need not be estimated as an entire simultaneous-equation system: consistent third-party projections and independently defended economic judgments may be used there. If you choose a different modeling architecture, explain why.

Distinguish a *systematic* model-based prediction from an *idiosyncratic* or judgmental adjustment. Explain what information motivates any override. Do not treat association as proof of causation. Show sufficient quantitative evidence—including uncertainty, competing specifications, diagnostic checks, or out-of-sample validation when appropriate—to make your conclusions credible.

Data and research documentation

Course sources may include BEA national-income and industry accounts, FRED and related macro series, as well as sources such as EARL and FactSet where available. Data availability and quality vary across economic concepts. Explain transformations, release lags, frequency changes, missingness, measurement choices, and data limitations when they materially affect results.

As the analysis evolves, record major research decisions in code, notebooks, README files, or other ordinary project documentation. A concise record should answer: **Question—Method—Evidence—Decision**. You need not prepare a separate weekly research essay or submit code for each interim presentation.

Presentation Schedule and Scope

Date	Presenting teams	Expected format and coverage	Points per team
Feb 17	A, B, C	One slide: macro, national, one industry	2
Feb 24	A, B, C	Updated one slide: macro, national, one industry	2
Mar 3	A, B, C	Updated one slide: macro, national, one industry	2
Mar 17	A	Full deck: macro, national, one industry	54 (UG) / 44 (GR)
Mar 24	B	Full deck: macro, national, two industries	54 (UG) / 44 (GR)
Mar 31	C	Full deck: macro, national, two industries	54 (UG) / 44 (GR)
Apr 7	A, B, C	Ungraded one-slide in-class update; connect toward company-level analysis	0
Final	A, B, C	Comprehensive TDPR; date and time on Canvas	30 (UG) / 25 (GR)

The calendar lists different industry coverage for Team A versus Teams B and C because the course and project evolve over time. Teams will be evaluated against the content and analytical expectations appropriate for their assigned presentation date; they will not be disadvantaged simply for presenting earlier.

One-Slide TDPR Updates

Purpose and format

Each team presents **one substantive slide** on February 17, February 24, and March 3. Plan for approximately **5–10 minutes** per team, including instructor and audience questions. One slide forces the team to select the most decision-relevant information and convey it clearly.

The slide should communicate the team's current macro outlook, associated national-income implications, at least one industry application, and the principal risk or uncertainty. It may use a compact forecast table, chart, scenario graphic, or other effective visualization. Cite major data sources and clearly date the forecast. Your conclusions may stay the same across updates; the evidence should be reassessed in light of new data and events.

Grading

Each of the three presentations is worth **two course points**. Evaluation emphasizes preparation, substantive and coherent analysis, supporting evidence, and the team's ability to discuss the slide. These are low-stakes cumulative updates, not separately graded written reports. The instructor does **not** grade underlying code at these stages. A student does not lose credit merely because a defensible forecast differs from the instructor's view.

The **April 7** one-slide discussion is an ungraded classroom activity. Use it to integrate new evidence and, as appropriate, begin connecting the macro, national, and industry analyses to company-level conclusions that will appear in the final submission.

Full-Deck TDPR Presentation

Format, timing, and audience

Every team leads exactly one major, graded presentation: **Team A on March 17, Team B on March 24, and Team C on March 31**. The team's allotted in-class time is approximately **45–60 minutes**, including questions and discussion. All members should prepare to explain and defend the team's research, even if speaking roles are uneven. A polished monologue is less valuable than an informative, responsive discussion.

Prepare a slide deck appropriate for a client with business experience but limited specialized knowledge of econometrics or ML. The slides should communicate the principal economic conclusions directly; technical specifications, diagnostic plots, and supporting results may be placed in an appendix for questions. Include attribution for substantive contributions, either by slide, section, or a concise summary.

Required coverage

- **Macro:** An NTM forecast for five macro KPIs, coherent economic rationale, and revisions from earlier expectations.
- **National income:** Quantified implications for major income-statement components, including revenues, costs, and profits.
- **Industry:** Team A discusses at least *one* industry; Teams B and C discuss at least *two* industries, consistent with the course calendar. Explain heterogeneity relative to national predictions.
- **Empirical evidence:** Show what the team actually estimated or tested; use appropriate quantitative support for material relationships, forecasts, or model choice.
- **Risk:** Discuss plausible alternatives, probability judgments, and material differences among scenarios. The scenarios should remain economically coherent across the layers of analysis.

The structure is intentionally flexible. A team may spend more time on a particularly consequential empirical issue and less time on material already established, provided its analysis remains complete, clear, and appropriately balanced.

Team evaluation

The team-level presentation is worth **54 course points for FECON 490** students and **44 course points for ECON 690** students. Evaluation considers three connected areas:

1. **Analytical rigor:** accuracy of economic and accounting logic; credible, internally consistent forecasts; appropriate empirical specifications; defensible inferences; and coherent treatment of scenarios and uncertainty.
2. **Evidence and materials:** quality, currency, and interpretation of data and model results; clarity of figures; appropriate source attribution; and professional organization.
3. **Presentation and defense:** direct communication to a business audience; effective time management; engagement with questions; and each student's demonstrated understanding.

Underlying scripts and data are tools the team should use throughout its research, but **they are not directly graded for the full-deck presentation**. The instructor will assess the empirical research through what the team presents and can explain. Code correctness and reproducibility receive direct assessment only with the final package.

Final Comprehensive TDPR

Role of the final submission

The final TDPR is the team's comprehensive, cumulative research product. It should integrate the best prior work, incorporate feedback, and extend the hierarchy through selected companies. Teams are not asked to repeat the earlier presentations verbatim; they should improve outdated assumptions, correct weaknesses, and clearly identify research decisions and major changes in outlook.

Required analytical content

1. **Macro:** NTM numerical forecasts and reasoned support for all five macro KPIs (real output, labor, prices, rates, and the dollar).
2. **National income:** NTM forecasts and justification for the national income statement, including revenues/gross output, intermediate inputs, costs, and relevant measures of profitability.
3. **Industries: Two industries not previously examined by the team** during the semester, drawn from **different sectors**. Quantify differences from national expectations and explain the drivers.
4. **Companies: One company within each selected industry**. Explain company-specific departures from industry projections, as well as implications for profitability and business decisions.
5. **Risk:** At least **three jointly specified economic scenarios** with defensible probabilities that sum to 100%. Follow the implications of each scenario from macro through national, industry, and company outcomes. Do not construct unrelated independent scenarios for every KPI.

6. **Methods and empirical evidence:** Document and evaluate the econometric research underpinning important predictions and the effect of judgmental adjustments. Include appropriate evidence of model assessment and forecast uncertainty. ML extensions are welcome when they contribute, but no fixed list of algorithms is required.

The final should provide an evidence-based account of what the team learned, including methods that proved unhelpful. A negative model-comparison result can be valuable if it is correctly implemented and thoughtfully interpreted.

Deliverables and submission

One student submits on behalf of the team by the date and time shown on Canvas. Provide:

1. **Recorded presentation:** An approximately **60-minute video** that communicates and defends the complete TDPR. Supply a viewable link (for example, an unlisted YouTube link with appropriate permissions), as directed on Canvas.
2. **Slide deck:** A PDF containing the final client presentation, with appropriate source citations and team attribution.
3. **Reproducibility archive:** A ZIP file containing the data, R or Python code, required dependency or environment information, and a README that explains how to reproduce material figures, tables, model estimates, and forecasts. Organize files logically and include sufficient comments for another technically competent analyst to understand the workflow. A Quarto-based workflow is acceptable; render a presentation-ready PDF with code hidden if helpful and include the source `.qmd`.
4. **AI Usage Log:** The team's cumulative `AI-Usage-Log.md`, described later in this document.
5. **Graduate-extension attribution (when applicable):** Identify each ECON 690 student's individual research question, analysis, code, evidence, and conclusions within the final materials or a clearly identified appendix. The proposal is an earlier, separately submitted approval step; a wholly independent final research report is not required.

Suggested filenames are `TDPR-TeamA-Final.pdf`, `TDPR-TeamA-Final.zip`, and analogous filenames for Teams B and C. The archive should include `README.md`, organized code and data folders, and the AI Usage Log. Submit the video link using the mechanism specified on Canvas. Exact platform instructions and deadlines will be posted there.

Final team-score rubric

The percentages below are *within the final TDPR assessment*; they do not represent additional course-grade categories.

Criterion	Share of final TDPR score
Macro analysis: forecasts, coherence, and rationale	15%
National income analysis: accounting links and evidence	15%
Industry analysis: two new industries and model support	15%
Company analysis: two companies and business implications	15%
Risk and scenario analysis: joint coherence and probabilities	15%
Code and reproducibility: correctness, execution, organization, documentation	15%
Professionalism and communication: presentation, slides, sources, clarity	10%
Total	**100%**

The final is worth **30 course points for undergraduate students** and **25 for graduate students**, prior to the individual-score process described below. The code component is the **only direct grading of code** in the semester's regular TDPR deliverables. Merely providing a large codebase is not sufficient: the submitted analysis must run, reproduce material outputs, and support the claims made in the presentation.

Individual Student Evaluation and Numeric Peer Review

Individual scores for major team deliverables

The full-deck presentation and final TDPR each receive a team score. Each student's score for those deliverables is composed of:

Element	Weight within individual deliverable score
Team score	25%
Numeric teammate peer review	25%
Instructor evaluation of individual contribution	50%
Total	**100%**

Numeric peer-review procedure

After each team's full-deck presentation, and again following the final submission, each student confidentially allocates **10 points per other teammate** among the other team members. Students cannot assign points to themselves. All allocated points must sum to the evaluator's prescribed budget; ties and equal allocations are permitted.

For example, with three students per team, each student has **20 points** to distribute across the other two team members. A student who believes both contributed equally allocates 10 points to each. Contributions need not be equal, and honest unequal assessments are appropriate when warranted.

If $p_{e,i}$ is the number of points evaluator e gives teammate i in a team of size N, the normalized peer score is

$$\text{PeerScore}_i = 100\,\frac{\sum_{e \neq i} p_{e,i}}{10(N-1)}.$$

Thus, if all teammates assign equal shares, each student receives a peer score of 100, independent of team size. The instructor may examine unusual allocations and corroborating evidence before determining individual scores. Numeric peer evaluations are not substitutes for the instructor's independent assessment.

Instructor evaluation and attribution

Instructor evaluation is informed by observed presentations and responses to questions, understanding of macroeconomic and business concepts, appropriate use and interpretation of econometric or ML evidence, and the quantity and quality of individually attributable contributions to research and team deliverables. At the final stage, documented contributions to code and its interpretation may also inform the instructor's evaluation; this does not create interim code grades.

Use unobtrusive attribution on slides (by slide or section), documented research ownership, code comments or authorship where appropriate, and other reasonable evidence to identify who contributed meaningfully. Attribution should be informative rather than bureaucratic. The three short presentations are two-point team updates; numeric peer-review surveys are reserved for the major full-deck and final milestones. These reviews, together with instructor observations over the semester, inform the major-deliverable individual scores.

Participation Activities

Macro Journal

The Macro Journal is a short, individual Canvas activity designed to give each student a regularly updated view of economic conditions before entering team discussions. Its established format asks whether the student's assessment of **output, labor-market health, and the price environment** has improved, deteriorated, or remained unchanged since the previous assessment, followed by a brief justification for each choice. Students may discuss economic information with classmates and may use permitted AI tools, but each submission must reflect the student's own reviewed understanding and reasoning.

The Journal falls within the **10-point participation category**, rather than a separate graded category.

Completion and a genuine effort to provide evidence-based justifications matter; no particular directional judgment is rewarded and final forecast accuracy is not graded. Canvas lists the relevant dates.

Audience survey

Non-presenting students complete a brief Canvas survey during each scheduled presentation session. The survey asks students to identify the main forecast or rating presented, compare selected conclusions to their own assessment, and briefly explain a point of agreement, disagreement, or uncertainty. For example, students may address an output or inflation view, a national-income implication, or an industry-level prediction. Presenters are exempt while their own team is presenting.

Surveys contribute to participation through thoughtful completion and active listening; they are **not** the same as confidential numeric teammate peer review. Audience questions and respectful discussion also contribute to participation.

Class exercises and April 7 activity

Hands-on exercises and the April 7 all-team update are **ungraded practice**. They provide opportunities to implement concepts, obtain feedback, and improve the TDPR. Preparedness and meaningful engagement form part of the overall participation assessment.

Graduate Research Extension (ECON 690)

Students enrolled in ECON 690 must complete an **individual Graduate Research Extension** tied directly to their team's TDPR. It is worth **15 course points** and is evaluated separately from the team's project score.

The extension is intended to show additional analytical depth, methodological judgment, and intellectual independence. It does *not* require the student to add another ML technique, duplicate the entire team's research, or produce an unrelated paper. A useful extension begins with an important uncertainty or methodological choice encountered in the team's ordinary work.

Possible research directions

- Compare alternative econometric or ML specifications for a macro-to-national, national-to-industry, or industry-to-company relationship.
- Evaluate stability across sample periods, recessions, inflation regimes, industries, or data revisions.
- Examine forecast accuracy with appropriate time-series splits, rolling windows, or other out-of-sample designs.
- Assess model uncertainty, coefficient uncertainty, prediction intervals, or scenario probabilities.
- Investigate aggregation, measurement, frequency, small-sample, leakage, or generated-regressor problems in the top-down hierarchy.
- Compare a baseline econometric model with a more flexible ML alternative and assess whether any improvement is practically meaningful.
- Pursue another substantive extension with prior instructor approval.

Research process

The work must address four explicit questions:

1. **Question:** What is the specific research issue, and why does it matter for the team's forecast or business decision?
2. **Method:** What additional analysis was performed, and why is it suited to the question?
3. **Evidence:** What empirical results, comparisons, diagnostics, or sensitivity analyses were generated?
4. **Conclusion:** What do the results imply for the team's TDPR, and what important limitations remain?

A carefully designed analysis concluding that the team should *not* change its method can receive full credit. The objective is independent insight, not a favorable result.

Proposal and final integration

Submit a brief, individual proposal describing the research question, motivation, planned method, and expected evidence. The instructor must approve the proposal before the extension proceeds. Due date and format will be provided on Canvas.

The extension may use and build on the team's data, code, and models, and its findings may appear naturally within the team's final presentation and reproducibility archive. The graduate student's contribution must be clearly and separately attributable; other teammates are not responsible for performing it. A separate full-length final research report is **not required** if the integrated materials clearly document the individual's question, method, evidence, conclusions, and code.

Extension evaluation

The individual extension will be evaluated for relevance and clarity of the research question, methodological rigor, correctness and reproducibility of implementation, quality of the evidence and validation, depth of interpretation and treatment of uncertainty, practical relevance to the TDPR, and evidence of individual contribution and professional documentation.

Artificial Intelligence Use

Approved services and appropriate uses

For this course, students may use **only Duke-provided AI services available to students in this course**, currently:

1. Microsoft Copilot: `https://oit.duke.edu/service/microsoft-copilot/`
2. ChatGPT Edu: `https://oit.duke.edu/service/chatgpt-edu/`

Other generative-AI services, including consumer versions of the same products, require explicit instructor approval. Ordinary search engines, statistics documentation, and non-AI research resources are not prohibited by this restriction.

Permitted services may assist with coding, debugging, explaining concepts, brainstorming, documentation, empirical model exploration, preparing presentations, and individual Macro Journals. However, students remain responsible for data choices, models, testing, interpretations, forecasts, probabilities, and all submitted work. AI-generated material must be understood, critically evaluated, checked against appropriate data or primary sources, tested where applicable, and corrected before use. It should **support** the student's reasoning rather than substitute for it. Do not give AI tools passwords, access tokens, restricted credentials, or confidential third-party information.

AI Usage Log

Each team must maintain a shared, cumulative `AI-Usage-Log.md` as part of its project files. Each student documents their own **substantive** AI interactions that materially affect project design, empirical modeling, coding, interpretation, or deliverables. Trivial spelling or formatting assistance need not be logged. The log should be updated during the semester and submitted for review with the final package; it is **not separately graded in the interim presentations**.

Each substantive entry should identify:

- **Date and student:** When the interaction happened and who used the service.
- **Milestone or file:** Which presentation, analysis, code file, or research question was affected.
- **Task and approved tool:** What was attempted and whether ChatGPT Edu or Microsoft Copilot was used.
- **AI assistance:** A concise summary of the prompt and resulting suggestion or contribution.
- **Verification and revisions:** How the student tested, checked, corrected, or rejected the output.
- **Outcome:** Whether the advice was adopted, revised, rejected, or retained for future investigation.

Illustrative entry: On February 3, Student A used Duke ChatGPT Edu while working on an R specification

for industry operating-cost growth. The service suggested a lagged regression and a cross-validation procedure. Student A inspected the date alignment, compared the results with a baseline, discovered leakage in the initial split, corrected it, and retained the revised specification only after verifying the out-of-sample calculations. A concise log entry would record the task, suggestion, error found, revision, and resulting decision.

The instructor evaluates responsible use through evidence of independent understanding, transparency, and thoughtful verification—*not* by the amount of AI used. The log supports the final research record and is evaluated only at the final stage, within the applicable final-submission criteria.

Research Attribution and Professional Standards

Provide a credible record of who developed key analyses, model specifications, economic assumptions, presentation materials, and code. Slide footnotes, section attribution, commits, code comments, and brief summaries are all acceptable mechanisms. Clearly identify research originally developed by outside analysts, the data and sources underlying figures, and relevant limitations of reused data or code. Work should be intelligible and reproducible without requiring the instructor to guess file dependencies or reconstruction steps.

Administrative Notes

- **Submission deadlines:** Dates and times for slides, the final video and archive, surveys, Macro Journals, and the graduate proposal are set on Canvas. The spreadsheet course calendar identifies the presentation sessions.
- **Team submissions:** One student may submit each team deliverable on the team's behalf. Every student remains responsible for meaningful participation and accurate individual attribution.
- **Judgment versus realization:** Neither an optimistic nor a pessimistic forecast receives credit by virtue of its direction. Support, logic, consistency, validation, and the quality of decisions are what matter.
- **Changes:** The instructor may update operational instructions in response to pedagogical or logistical needs, with notice through Canvas.